Ex 6.2.2

Suppose that D is simply connected and that $f \in H(D)$ has no zeros in D. Show that there is an analytic branch of $\log(f(z))$ in D; that is, show there is a $g \in H(D)$ so that $f(z) = \exp(g(z))$ for all $z \in D$. (Hint: $f'/f \in H(D)$ and must have an anti-derivative.)

Pf. From the hint, since f is nonvanishing on D, we have from Prop 2.2 (Quotient Rule) that the logarithmic derivative f'/f is holomorphic on D.

Since D is simply connected, Thm 6.17 (Cauchy for Simply Connected Domains) gives that every function in $H(D)$ has an antiderivative on D. Applying this to $f'/f \in H(D)$, $\exists F \in H(D)$ s.t.

$$F'(z) = \frac{f'(z)}{f(z)}$$

$\forall z \in D$. Let $h(z) := f(z)\exp(-F(z))$ on D. Both factors are holomorphic, from Prop 2.16 where $\exp \in H(\mathbb{C})$ and Prop 2.2 (Chain Rule). Then $h \in H(D)$. By the product and chain rules, we have,

$$\begin{aligned} h'(z) &= f'(z)\exp(-F(z)) - f(z)F'(z)\exp(-F(z)) \\ &= \exp(-F(z))\left(f'(z) - f(z)\tfrac{f'(z)}{f(z)}\right) \\ &= 0 \end{aligned}$$

$\forall z \in D$. Since D is a domain and $h' \equiv 0$, we have from Ex 2.2.2 that this forces h to be constant on D, say $h \equiv c$. Because f never vanishes and exp is nonzero everywhere, $c \neq 0$.

We have that the complex exponential is surjective onto $\mathbb{C} \setminus \{0\}$ (from Prop 2.16 and Def 1.24), so we can take $\alpha \in \mathbb{C}$ with $e^{\alpha} = c$.

Let $g(z) := F(z) + \alpha$. Then $g \in H(D)$, and $\forall z \in D$, we have,

$$\begin{aligned} f(z) &= h(z)\exp(F(z)) \\ &= c\,\exp(F(z)) \\ &= \exp(\alpha)\exp(F(z)) \\ &= \exp(F(z) + \alpha) \\ &= \exp(g(z)) \end{aligned}$$

Therefore, we have that g is the required analytic branch of $\log(f(z))$ on D. □

Ex 6.2.4

Show that the polynomial $p(z) = z^5 + 3z^3 + 7$ has five distinct zeros inside $|z| = 2$. (Use Ex 6.2.3.)

Pf. Count the zeros of p inside $|z| = 2$ following Ex 6.2.3 (Rouché's theorem). Then we can split p as $f + h$ with, $f(z) = z^5$ and $h(z) = 3z^3 + 7$.

Therefore, $p = f + h$ and $p - f = h$. On the closed disk $\overline{D_2(0)}$, f and p are entire (i.e. holomorphic on an open neighborhood of $\overline{D_2(0)}$). On the circle $|z| = 2$ we can say,

$$|p(z) - f(z)| = |h(z)| \leqslant 3|z|^3 + 7 = 24 + 7 = 31 < 32 = 2^5 = |f(z)|$$

Therefore, we have Ex 6.2.3's condition of $|f - p| < |f|$ on $|z| = 2$. Hence, f and p have the same number of zeros inside the disk $|z| < 2$, counted with multiplicity.
Since $f(z) = z^5$ has a single zero of order 5 at the origin, p has exactly five zeros in $|z| < 2$ when multiplicities are included.

By Lemm 5.4 (factorization of holomorphic functions at a zero of finite order), a zero a of p has order $m \geqslant 1$ iff $p(a) = p'(a) = \cdots = p^{(m-1)}(a) = 0$ and $p^{(m)}(a) \neq 0$. Therefore, every zero is simple precisely when p and p' don't share zeros. We can evaulate,

$$p'(z) = 5z^4 + 9z^2 = z^2\left(5z^2 + 9\right)$$

Hence $p'(z) = 0$ iff $z = 0$ or $z^2 = -9/5$. Then we can consider if:

- $\underline{z = 0}$: $p(0) = 7 \neq 0$, so 0 is not a zero of p.
- $\underline{z^2 = -9/5}$: z is purely imaginary and nonzero. Using $z^2 = -9/5$ to reduce $p(z)$, we have,

  $$z^3 = zz^2 = -\tfrac{9}{5}z$$
  $$z^5 = z(z^2)^2 = \tfrac{81}{25}z$$

  Therefore,

  $$p(z) = \tfrac{81}{25}z + 3\left(-\tfrac{9}{5}z\right) + 7 = \left(\tfrac{81}{25} - \tfrac{135}{25}\right)z + 7 = -\tfrac{54}{25}z + 7$$

  Let $p(z) = 0$, this gives $z = \frac{175}{54} \in \mathbb{R}_{>0}$, contradicting that z is purely imaginary and nonzero. Therefore, we have that no common zero exists.

Therefore p and p' share no zero in $\mathbb{C}$, and every zero of p has order 1. Using the count above, p has exactly five distinct zeros inside $|z| = 2$. □

Ex 6.2.6

Suppose that D is open and u is harmonic on D. Show that u must be smooth in the sense that it has continuous partials of all orders in D.

Pf. Since the smoothness of u is a local, we need to show u has continuous partials of all orders in a neighborhood of each point of D.

Take $a \in D$. Since D is open, we take $r > 0$ with $D_r(a) \subseteq D$. Then the open disk $D_r(a)$ is convex, hence simply connected using Lemm 6.12.
On a simply connected domain, every harmonic function has a harmonic conjugate from Thm 7.13(h) (Simply Connected Domains). This gives a harmonic function v on $D_r(a)$ s.t. $f(z) := u(z) + iv(z) \in H(D_r(a))$.

Since f is holomorphic on $D_r(a)$, this means that the components $u = \mathrm{Re}(f)$ and $v = \mathrm{Im}(f)$ have to be smooth using Ex 4.3.4, i.e. if $f \in H(D)$, then both u and v have continuous partial derivatives of all orders on D.
In particular, u has continuous partials of all orders on the neighborhood $D_r(a)$ of a.

Since we took any $a \in D$, u has continuous partials of all orders at every point of D, as required. □

Ex 7.1.3

Let D be a domain and (f_n) a sequence in $H(D)$ that converges pointwise to f on D. Show that if $\mathcal{F} = \{f_n : n \in \mathbb{N}\}$ is a normal family, then (f_n) converges to f uniformly on compact subsets of D. In particular, $f \in H(D)$.

Pf. From Def 7.4, we know that a family $\mathcal{F} \subseteq H(D)$ is normal if every sequence in $\mathcal{F}$ has a subsequence converging uniformly on compact subsets of D to some limit in $H(D)$. The limit of any such convergent subsequence has to be the pointwise limit f.

Let $K \subseteq D$ be compact. We can show $f_n \to f$ uniformly on K using proof by contradiction. Assume, for proof by contradiction, that there exist $\varepsilon > 0$ and indices $n_1 < n_2 < \cdots$ s.t.

$$\sup_{z \in K} \left| f_{n_k}(z) - f(z) \right| \geqslant \varepsilon \tag{$*$}$$

$\forall k \in \mathbb{N}$. Then the subsequence $(f_{n_k}) \subseteq \mathcal{F}$ lies in the normal family $\mathcal{F}$. Applying Def 7.4 to this, we have that a further subsequence $(f_{n_{k_j}})$ converges uniformly on compact subsets of D to some $g \in H(D)$.

From uniform convergence we have pointwise convergence, so $\forall z \in D$, we have,

$$g(z) = \lim_{j \to \infty} f_{n_{k_j}}(z) = \lim_{n \to \infty} f_n(z) = f(z)$$

using (f_n) converges pointwise to f. Hence $g = f$ on D. In particular, $f = g \in H(D)$.

But we had that $f_{n_{k_j}} \to f$ uniformly on the compact set K, so $\sup_K |f_{n_{k_j}} - f| \to 0$, which contradicts $(*)$. Therefore $f_n \to f$ uniformly on every compact $K \subseteq D$, and $f \in H(D)$. □

Ex 7.2.2

Suppose that $f : \mathbb{C} \to D$ is a conformal equivalence. Show that $D = \mathbb{C}$ and that $f(z) = az + b$ with $a \neq 0$.

Pf. Using Rmk 5.51, i.e. a conformal equivalence $f : \mathbb{C} \to D$ is a holomorphic bijection whose inverse is holomorphic. Using Cor 5.49, this is trivial when f is a holomorphic bijection. In particular, f is entire and one-to-one.

We can use the solution of Ex 5.5.6 last week, where we considered the singularity of $g(z) = f(1/z)$ at 0. We ruled out the removable case using Thm 5.26 (Liouville), essential using Thm 5.21 (Casorati–Weierstrass) and Thm 5.38 (Open Mapping Theorem), and used Lemm 5.2.3 to deduce that this is a pole. We have that entire injective function has the form $f(z) = az + b$, where $a \neq 0,\ b \in \mathbb{C}$.

An affine map of this type is surjective onto $\mathbb{C}$, therefore $D = f(\mathbb{C}) = \{az + b : z \in \mathbb{C}\} = \mathbb{C}$. □

Ex 7.4.2

Suppose that D is an open connected subset of $\mathbb{C}$.

(a) If $p \in D$, show that $D \setminus \{p\}$ is connected. (You can assume that any deleted neighborhood, $D_\varepsilon(p)$, is path connected and therefore connected.)

(b) Show that if D is a neighborhood of ∞, then $D \cup \{\infty\}$ is connected in $\mathbb{C}^+$.

Pf.

(a) Since D is a domain: from Ex 2.1.5, we have that it is path connected. Using Lemm 6.1.2, any convex set is path connected, so we also have that every open disk in D is also path connected.

Using the fact that that path connected sets are connected (from Sec 2.1), we take a basepoint $z_0 \in D \setminus \{p\}$, and take any $z \in D \setminus \{p\}$. We take a continuous path $\gamma : [0,1] \to D$ with $\gamma(0) = z_0$ and $\gamma(1) = z$. Then we can consider if:

- $p \notin \gamma([0,1])$: γ is already a path in $D \setminus \{p\}$ from z_0 to z.
- $p \in \gamma([0,1])$: The preimage,
$$E := \gamma^{-1}(\{p\}) = \{t \in [0,1] : \gamma(t) = p\}$$
is nonempty. Using continuity of γ, it also closed in $[0,1]$, hence compact. Let $t_{\min} = \min E$ and $t_{\max} = \max E$.
Since $\gamma(0) = z_0 \neq p$ and $\gamma(1) = z \neq p$, we have $0 < t_{\min} \leqslant t_{\max} < 1$.
Take $\delta > 0$ s.t. $D_\delta(p) \subseteq D$, which is possible since D is open. Using the hint, the punctured disk $D_\delta(p) \setminus \{p\}$ is path connected.
By continuity of γ at $t_{\min}$ and $t_{\max}$, $\exists \varepsilon > 0$ s.t. $0 < t_{\min} - \varepsilon$, $t_{\max} + \varepsilon < 1$, and $|\gamma(t_{\min} - \varepsilon) - p| < \delta$, $|\gamma(t_{\max} + \varepsilon) - p| < \delta$.
We can also take $t_{\min}$ and $t_{\max}$ as the extremes of E, which gives $\gamma(t_{\min} - \varepsilon) \neq p$ and $\gamma(t_{\max} + \varepsilon) \neq p$, i.e. both endpoints lie in $D_\delta(p) \setminus \{p\}$. Now we can take a path η from $\gamma(t_{\min} - \varepsilon)$ to $\gamma(t_{\max} + \varepsilon)$ inside $D_\delta(p) \setminus \{p\} \subseteq D \setminus \{p\}$. Then we have the concatenated path,
$$\gamma\big|_{[0,t_{\min}-\varepsilon]} \;\; \eta \;\; \gamma\big|_{[t_{\max}+\varepsilon,1]}$$
is a continuous path from z_0 to z entirely inside $D \setminus \{p\}$. The two outer pieces avoid p using $t_{\min}$ and $t_{\max}$, and the middle piece η avoids p also.

In either case z_0 and z are joined by a path in $D \setminus \{p\}$, so $D \setminus \{p\}$ is path connected, hence connected. □

(b) We have $\exists R > 0$ s.t. $V := \{z \in \mathbb{C} : |z| > R\} \subseteq D$ using the form of neighborhoods of ∞ in Lemm 5.23. Take $W := V \cup \{\infty\} \subseteq \mathbb{C}^+$.

We have that the inversion $T : \mathbb{C}^+ \to \mathbb{C}^+$ given by $T(z) = 1/z$ for $z \neq 0, \infty$ together with $T(0) = \infty$ and $T(\infty) = 0$ is a homeomorphism. Therefore with T,
$$T(W) = T\big(\{\infty\} \cup \{|z| > R\}\big) = \{0\} \cup \{|w| < 1/R\} = D_{1/R}(0)$$
which is an open disk, hence connected (convex). Therefore W is connected.

We have $D \cup \{\infty\} = D \cup W$ since, $V \subseteq D$ gives $W = V \cup \{\infty\} \subseteq D \cup \{\infty\}$, so $D \cup W \subseteq D \cup \{\infty\}$. The reverse inclusion is immediate since $\{\infty\} \subseteq W$.

Both D and W are connected subsets of $\mathbb{C}^+$, and they overlap: any point of V, e.g. $2R$, lies in both. Therefore $D \cup W$ is connected (union of two connected sets with nonempty intersection). By above, $D \cup \{\infty\}$ is shown to be connected in $\mathbb{C}^+$. □